

Classical Mechanics

Westlake University

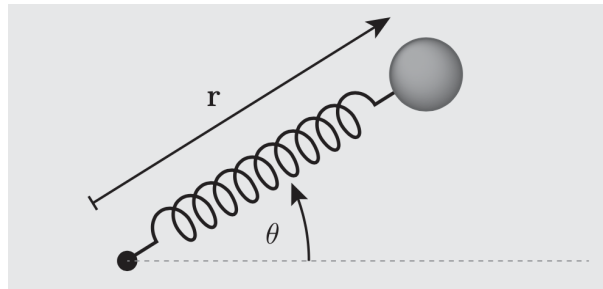
Total Points: 160 points

Deadline: 9:00 pm, September 22nd, 2025

All problem sets and your solutions must be kept private. Please do not share them publicly or with future students. All answers must be **handwritten**, then scanned or photographed, and submitted via Canvas.

A particle in two dimensions attached to a spring [20 points]

A block of mass m is free to move on a frictionless tabletop under the influence of an attractive Hooke's-law spring force $F = -k\mathbf{r}$, where the vector $\mathbf{r}$ is the position vector of the particle measured from the origin. Please derive the motion $x(t)$, $y(t)$ of the ball and show that the angular momentum of the ball about the origin is conserved.



Projectile with drag [15 points]

Consider a projectile subject to a drag force $\mathbf{F} = -m\alpha\mathbf{v}$. If it is fired with speed v_0 at an angle θ , it can be shown that the height as a function of time is given by

$$y(t) = \frac{1}{\alpha} \left(v_0 \sin \theta + \frac{g}{\alpha} \right) (1 - e^{-\alpha t}) - \frac{gt}{\alpha}. \quad (1)$$

Show that this reduces to the usual projectile expression $y(t) = v_0(\sin \theta)t - gt^2/2$. What exactly is meant by “small α ”?

Bonus question [5 points]: Derive Equation (1) using Newton’s second law.

Potential energies for positive power-law forces [15 points]

A particle moves in one dimension subject to the power-law force $F = -kx^n$, where the coefficient k is positive, and n is a positive integer. Please find the potential energy of the particle and also the maximum distance $x_{\max}$ it can reach from the origin, in terms of its maximum speed $v_{\max}$. The maximum distance is the **turning point** of the particle, because as the particle approaches this position it slows down, stops at $x_{\max}$, and turns around and heads in the opposite direction.

Problems from Classical Mechanics of John R. Taylor [110 points]

Chapter 1: 1.27, 1.46, 1.37, 1.47 [$10 \times 3 + 15$ points]

Chapter 4: 4.2, 4.3, 4.21, 4.23, 4.27, 4.25 [$10 \times 5 + 15$ points]